

OMER GURSOY

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EDUCATION

Carnegie Mellon University

Expected May 2027

B.S. Information Systems, Minor in Machine Learning

Pittsburgh, PA

- **Relevant Coursework:** Machine Learning, Managing Decisions, Designing Human Centered Software, Intro to Entrepreneurship, Database Design and Development, Cognitive Psychology of Decision Making, Data Structures and Algorithms, Probability Theory, Matrix Algebra, Discrete Mathematics

TECHNICAL SKILLS

Programming Languages: Python, C, Javascript, HTML, CSS, Golang, SQL, LaTeX

Tools: Excel, PowerPoint, GitHub, PHPMyAdmin, MLFlow

WORK EXPERIENCE

Lead Software System Architect and Developer

May 2025 – Ongoing

DevSecOps Masters Professor at CMU - DevSecOps

Pittsburgh, PA

- Currently creating an AI Bill of Materials (AI-BOM) under the Linux Foundation SPDX 3.0 repository
- Making API calls to MLFlow to retrieve necessary data, using as input for a black-box model which outputs a json file to clients
- An AI-BOM is automatically updated when clients alter entries

Information Systems Consulting Intern

Jul 2016 - Present

Sevan Bicakci Couture Jewellery Atelier - Management Team

Istanbul, Türkiye

- Digitised legacy records by extracting data to excel, supporting the transition into CANIAS ERP for building a digital database
- Facilitated collaboration between technical and business teams, enabling the digitisation of 1500+ legacy records for ERP migration
- Developed a 4-year ML roadmap optimising inventory management - increasing the profits by 30%
- Met up with clients in London and conducted numerous sales in the Istanbul branch

Software Engineering Intern

Jul. 2024 – Aug 2024

NGN - Cloud Engineering Team

Istanbul, Türkiye

- Deployed and secured 5 servers in the cloud against malicious hackers
- Set up Greenbone Vulnerability Management Software using Docker
- Conducted vulnerability scans on multiple Ubuntu Servers, reporting 15+ security issues
- Gained hands-on experience in system security through 50+ hours of training on networking and low-level systems

Data Science Intern

Jul 2022 – Aug 2022

Tazi AI Systems - Data Science Team

Pittsburgh, PA

- Delivered data analysis code used in Machine Learning algorithms via use of the Pandas Python Library
- Assessed communications department with English script translation of video content

PUBLICATIONS

Omer Gursoy "The Crucial Role of Computer Science in Couture Jewellery Business and Theft Detection"

Istanbul Chamber of Jewellery

Oct 2022

PROJECT EXPERIENCE

Quantitative Analysis Algorithm | Python, Github, Web-Scraping, Prediction Algorithms

Apr 2023

- Implemented quantitative analysis algorithm in Python based on the Black-Scholes model within 14 hours
- Utilized web-scraping, Y-finance API, to find optimal stock ticker call price based on time until maturity
- Designed and built UX interface for stock prediction display using Matplotlib

Say it With Sweets | Javascript, HTML, CSS

Jan 2024 - Feb 2024

- Redesigned and Implemented web portfolio for *Say It With Sweets*, a Pennsylvania based bakery startup
- Created drop-down menus, custom color palettes, and improved UI/UX features for staff of 10+

CoVM | C, Co

Apr 2025

- Implemented a virtual machine for Co (C-like language developed at CMU) that can interpret and execute bytecode
- Influenced by the Java Virtual Machine and a low-level virtual machine for compiler backends

ACTIVITIES, SOCIETIES, AND INTERESTS

Interests: MMA, Public Speaking, Philosophy, Entrepreneurship, Fencing, Soccer, Tennis

Societies: Muslim Students Association, MMA Club, Spanish and Latin Student Association, Soccer Club